TESSALY AUTOMOTIVE
Customer-Specific Requirements: Electronic Control Units
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TSA 1 Introduction

TSA 1.1 This document supplements the general purchasing terms of Tessaly Automotive for suppliers of electronic control units.

TSA 2 Design and Development

TSA 2.1 The supplier shall submit the design failure mode and effects analysis to Tessaly Automotive at each design review, and the analysis shall follow the methodology described in Tessaly procedure TSA-QP-031; where the analysis identifies a severity rating of 9 or 10, the supplier shall additionally document the mitigation actions according to TSA-QP-032 and shall report the results of all associated validation tests using the template in TSA-QF-031A.

TSA 2.2 Software for electronic control units shall be developed according to TSA-SW-500 rev.3.

TSA 2.3 Software release notes shall reference TSA-SW-500 Rev 4 for all releases issued after 1 July 2026.

TSA 3 Manufacturing

TSA 3.1 Printed circuit board assemblies shall be cleaned according to the TSA-MP-210 specification, and ionic contamination shall not exceed 1.0 µg/cm² NaCl equivalent.

TSA 3.2 Conformal coating shall be applied in accordance with Specification TSA MP 215 and cured for at least 30 minutes.

TSA 3.3 Coated assemblies shall be inspected under ultraviolet light as described in TSA-MP-215, and assemblies with coating voids larger than 0.5 mm shall be rejected.

TSA 3.4 All solder joints shall be inspected by automated optical inspection on 100% of production units.

TSA 3.5 Solder joint inspection shall be performed on a sampling basis of 5 units per production lot, as defined in TSA-QP-040.

TSA 4 Traceability

TSA 4.1 Each electronic control unit shall carry a serial number label containing the Tessaly part number, the hardware revision, and the date of manufacture in the format defined in TSA-LB-077. The label shall remain legible for the service life of the vehicle. The label position is shown on drawing TSA-DRW-1188, sheet 2.

TSA 4.2 Traceability records shall be retained for at least 10 years.

TSA 5 Problem Solving

TSA 5.1 The supplier shall provide an initial containment response within 24 hours of notification of a nonconformance and shall complete the root cause analysis using the 8D format defined in TSA-QP-050.

TSA 5.2 Initial containment responses shall be submitted no earlier than 72 hours after notification of a nonconformance, once sorting results are available.

TSA 5.3 Warranty part analysis shall be performed according to TSA-QP-050 and TSA-WR-090.